

Constraint-Based Generation With Diffusion Model Using MCMC Sampling

Link for the Report

https://docs.google.com/document/d/1-XDswSFPv_BXPr9iCwxYoq17dw7MCf9zzZQKOEDbXTw/edit?usp=sharing

Team Members

22M0772	Piyush Sawarkar
213050027	Sakharam Gawade
190100027	Ashutosh Sharma
214057001	Saiful Haq

Problem Statement

There have been many recent works for conditional generation using various generative models like GANs and Diffusion Models, wherein instead of passing a uniform/normally distributed input to the model, they directly input the image/object to condition on as input.

We aim to solve a generation problem, wherein instead of an image/object to condition on, we have a constraint we would like the generated object to follow.

Our approach requires a continuous user-generated function that scores the input image, giving a higher score if the constraint is followed and vice-versa, allowing for gradients to backprop.

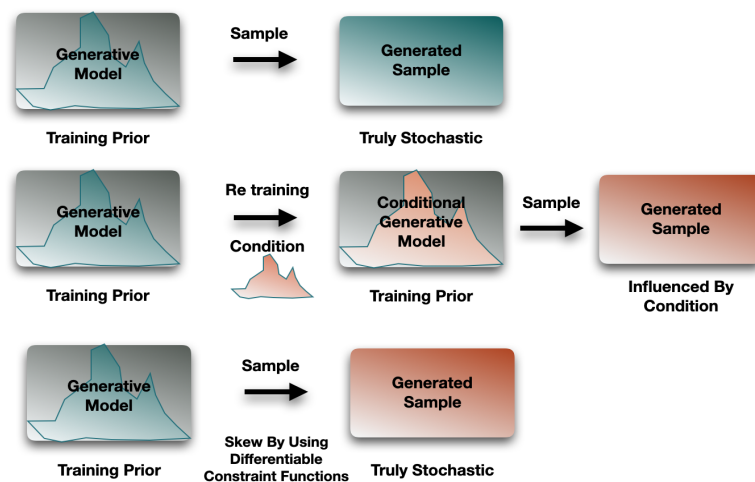
We decided to use sampling techniques, because, for a model to follow a specific constraint, we do not want to retrain the model from scratch, and want to use a well-trained model. Thus requiring no training in the diffusion model.

Motivation

Whenever one tries to sample from a Generative model for instance Diffusion model, then it will try to produce samples that are truly random and that are coming straight from the distribution on which it was trained.

So now if we want to generate something of our desire and **NOT** at random then one possible approach one could think of is resorting to Conditional Generative Models like the Conditional Diffusion Model, which would basically take an instance as input and conditioned on that would produce an output. But the problem with this approach is that we have to train such models first, which would totally make our original generative model **useless** as **re-training** is involved in the alternate approach.

Then One can ask an obvious question, Can I use 'somehow' the trained model I have so as to generate the right desired outcome similar to in the case of conditional generative models and NOT at random, but **without** training? So this is what our endeavor in this Project is all about.



Literature Survey

In recent years, there has been a growing interest in image generation and manipulation using deep learning techniques. Several approaches have been proposed for various image generation tasks, such as image-to-image translation, inpainting, and text-to-image synthesis. In this literature survey, we summarize five recent papers that propose novel methods for these tasks using denoising diffusion probabilistic models (DDPM) and generative adversarial networks (GAN).

"UNIT-DDPM: Unpaired Image Translation with Denoising Diffusion Probabilistic Models," introduces a new method for unpaired image-to-image translation using DDPM. The method achieves state-of-the-art performance without the need for adversarial training and generates high-quality images using a denoising Markov Chain Monte Carlo approach.

"Constrained Generative Adversarial Networks for Interactive Image Generation," proposes a GAN framework that allows humans to be "in-the-loop" of the image generation process by iteratively accepting relative constraints. The feedback from users is used to constrain the output of the GAN with respect to an underlying semantic space, enabling interactive control over image generation without sacrificing image quality.

"RePaint: Inpainting using Denoising Diffusion Probabilistic Models," presents an inpainting approach based on a DDPM that is capable of generating high-quality and diverse output images for any inpainting form. RePaint outperforms state-of-the-art autoregressive and GAN approaches by employing a pretrained unconditional DDPM as the generative prior and only altering the reverse diffusion iterations by sampling the unmasked regions using the given image information.

"Arbitrary Style Guidance for Enhanced Diffusion-Based Text-to-Image Generation," proposes a style guidance method for generating high-quality and diverse images using diffusion-based text-to-image generation models. The method can be guided by a reference image to specify fine-grained style features such as color distributions and brush strokes, and does not require a separate style transfer model.

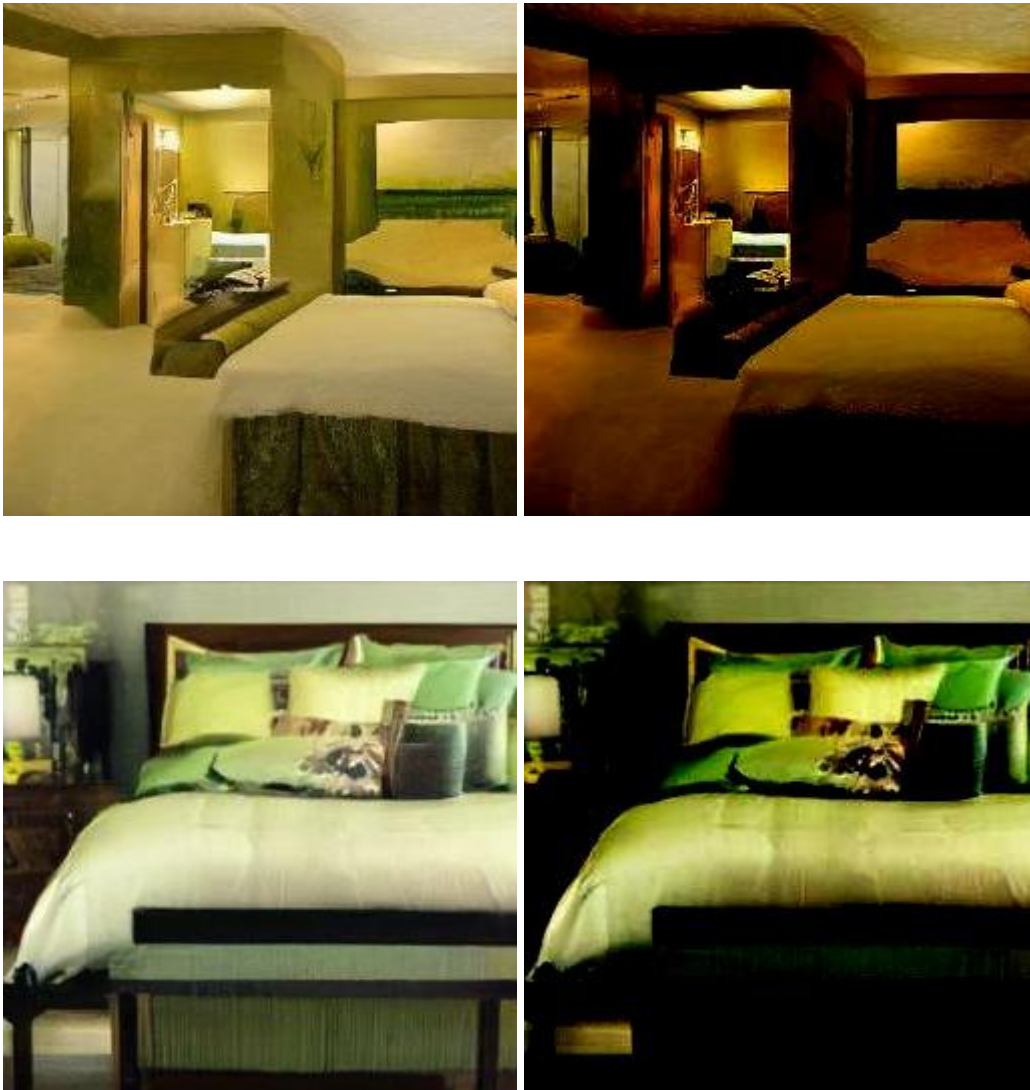
"More Control for Free! Image Synthesis with Semantic Diffusion Guidance," introduces a method for fine-grained, continuous control of DDPM for controllable image synthesis using a unified framework for semantic diffusion guidance. This allows for language or image guidance, or both, without re-training the model, and can be applied to datasets without associated text annotations.

Approach 1

Experiment Details

- 1) In the diffusion.py, for our experiments, we fixed the function to be the one measuring the amount of redness in the image.
- 2) The function is given as $F = \text{Sum}[(\text{Red}) - 0.5 * (\text{Blue} + \text{Green})]$
- 3) We used Langevin Metropolis Hasting Sampling at each denoising step.
- 4) The input to the scoring function for metropolis hasting was set to the image x_t .

Results



Analysis

On some range of random seeds, the output image changes visibly, but for most of the random seeds, there is no visible change (although the score of image increases in 3rd decimal place).

Approach 2A

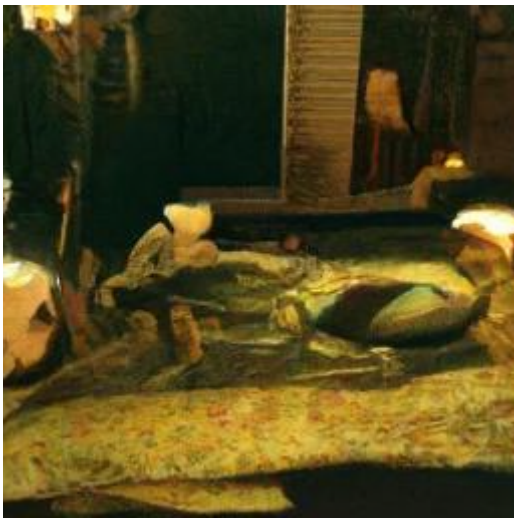
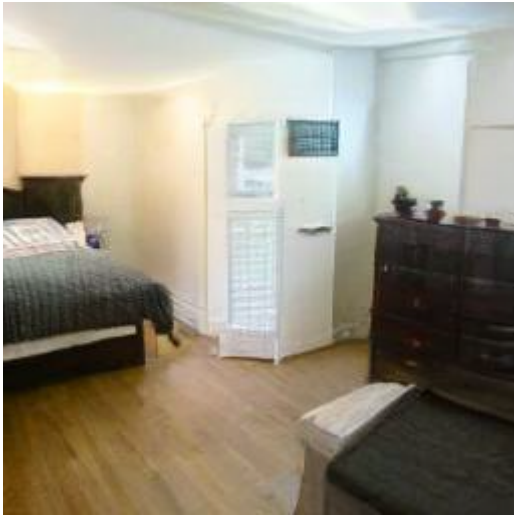
Use denoised image x_0 instead of x_t at each step for using MCMC updates

Experiment Details

- 1) Rather than using MCMC on image x_t , we first get x_0 from x_t by using diffusion denoiser, and use mcmc by calculating score gradient on x_0 calculated and update x_t
- 2) On following above methodology, when we did mcmc updates on each denoising step, running time was around 1 day, so the tau values we checked on gave gaussian noise output.
- 3) But, surprisingly, when we did mcmc steps on subsampled step, let's say every 100 denosiing steps, we got some results
- 4) But, on changing the scoring function, to let's say increase greenness of the image, we get the same resulting image, with possibly few minor changes (leading to change in scoring function in decimal places), bu visibly the same image

Results





Analysis

On using denoising module of the diffusion model for getting the correct underlying image for mcmc update, surprisingly on doing mcmc update at each step led to bad performance (image is totally noisy, no underlying image visible), but on aking mcmc step every 100 iterations, image changes to give better score (changes in 2nd decimal place, score maximum is 1)

Approach 2B

Use a denoised image for taking the MCMC step
Along with Gaussian Blur

Experiment Details

Rather than going for the denoised version x_0 from x_T , we could generate the denoised image from the x_T itself and proceed with the routine MCMC procedure in the reverse diffusion step.

Results



Analysis

On using gaussian blur for getting denoised image for mcmc update,

similar observations were there as approach 1, partly due to the fact that gaussian blur cannot retrieve underlying image from a very noisy image in the initial steps of the diffusion process.

Combined Analysis of Approach 1, 2A, 2B

Approach 1: Changing the random seeds causes some visible changes in the output image, but most random seeds do not show any visible changes except for a small increase in the image score.

Approach 2b: Using a Gaussian blur to denoise the image for MCMC updates also shows similar observations as approach 1, as the blur cannot fully retrieve the underlying image in the initial stages of the diffusion process.

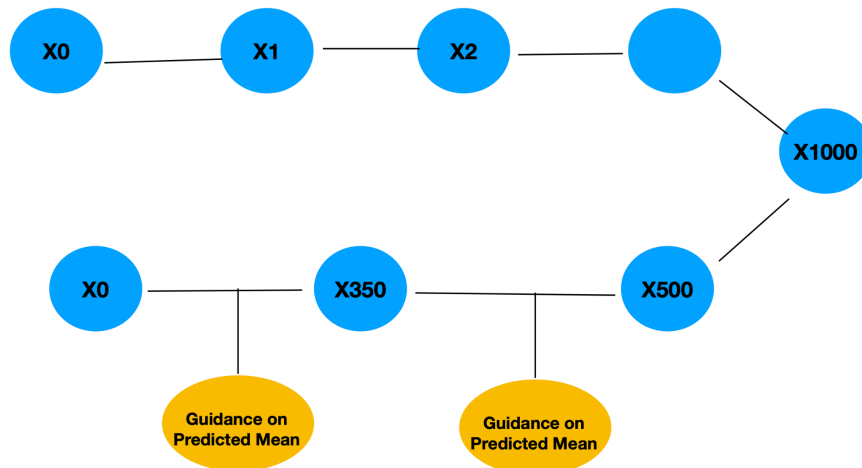
Approach 2a: Using a denoising module in the diffusion model to obtain the correct underlying image for MCMC updates surprisingly performs poorly when updates are made at each step, leading to a totally noisy image. However, when MCMC updates are made every 100 iterations, the image improves significantly with changes in the 2nd decimal place of the score.

When a random seed is fixed, the generated image from approach 2a does not change when the function is switched from increasing redness to blue/green. This suggests that fixing the random seed fixes the output distribution such that only a few modes remain, and the remaining denoising steps bring the image closer to the mode of the output distribution despite the attempts of MCMC updates to bring it closer to the constraint.

Approach 3

Guidance signal on predicted mean

Goal: We want to control the colour of the objects (Red, Blue, Green) in the image generated using the reverse process in Diffusion.



Motivation

Existing approaches use guidance signals on the predicted mean to control image generation, however the guidance function needs to be trained on the Noisy data. We want to introduce a method that can generate target images without the need of model training.

Experiment Details

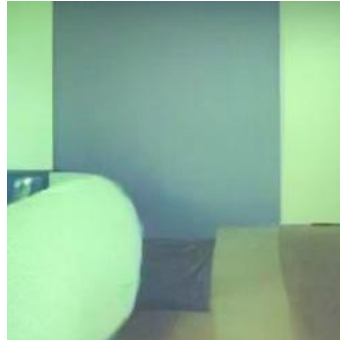
1. After N steps, we predict mean and mean_diff .
 - a. Mean is predicted using X , where X is the denoised image at the previous step.
 - b. Mean_diff is predicted using $X_{\text{original}} - X_{\text{original}}[\text{if channel} \neq \text{target channel, then pixel} = 0]$
2. $\text{Mean} = \text{Mean} - \text{scaling_factor} \times (\text{Mean} - \text{Mean_diff})$

Here, N and scaling factor are hyper-parameters.

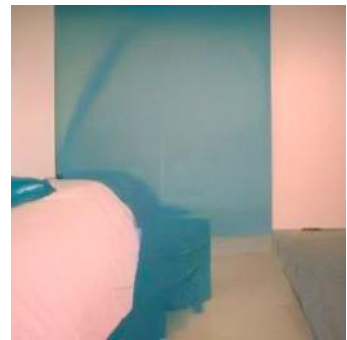
Results



Input Image



Green Output



Pink Output

Analysis

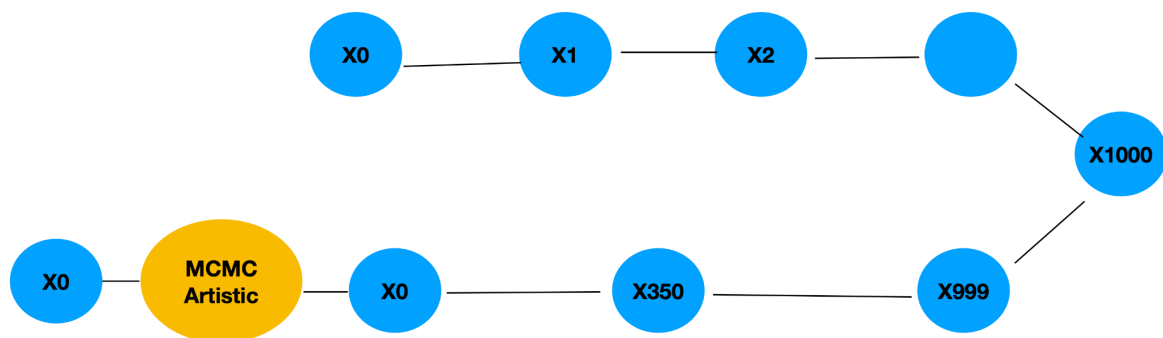
In both experiments, $N = 500$ and scaling factor = 0.001. We observe that the output adapts to the target colours however there is a slight difference (the painting is missing on the top left in the output).

Approach 4A

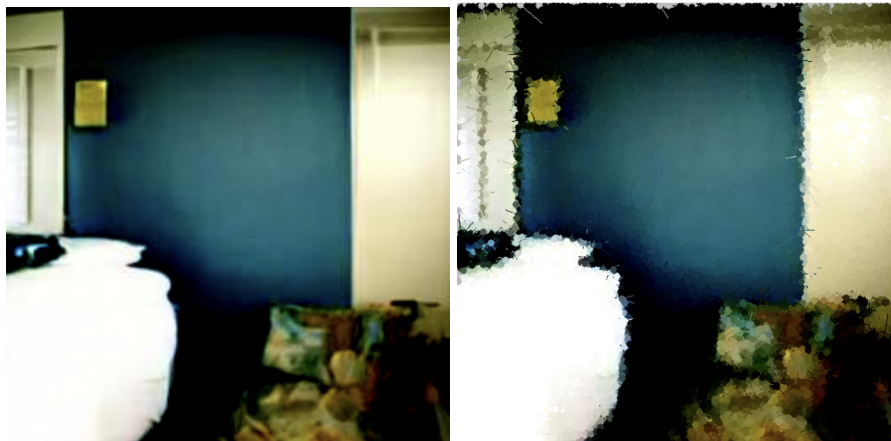
Diffusion Model + MCMC Sampling (Artistic) Sequential Algorithm

Experiment Details

In this approach, we apply an MCMC step after an image is fully sampled via the reverse diffusions step. This basically is the naive approach in which we are trying to skew the output distribution towards the desired one in a sequential fashion, first Diffusion unit then MCMC Unit. We use <https://aday651.github.io/posts/2022/02/mcmc-images/> for sampling artistic images.



Results



Original Image

Artistic Image (Band=10)

Analysis

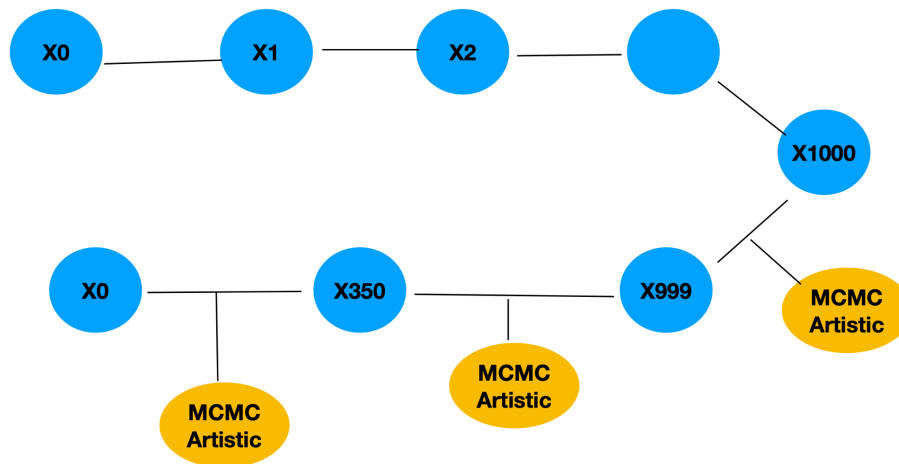
The artistic image looks like an oilpainting of the original image.

Approach 4B

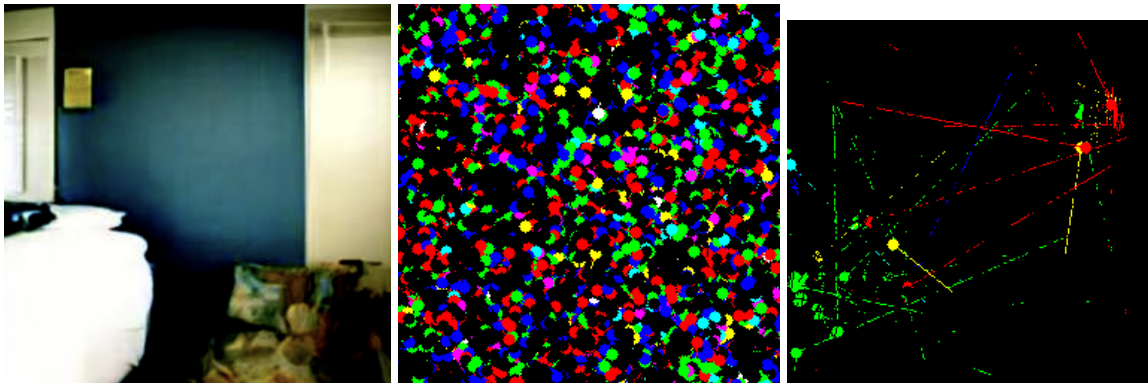
Diffusion Model + MCMC Sampling (Artistic)

Experiment Details

This is a little more mature approach as compared to previous where, we apply an MCMC step not after an image is fully sampled via the reverse diffusions step, but basically between every consecutive step of the reverse generation process. Here we expect to have similar outcome as the previous one although this should be a bit faster than the previous baseline code.



Results on Band=10



Original Image

100th Step

1000th Step

Analysis

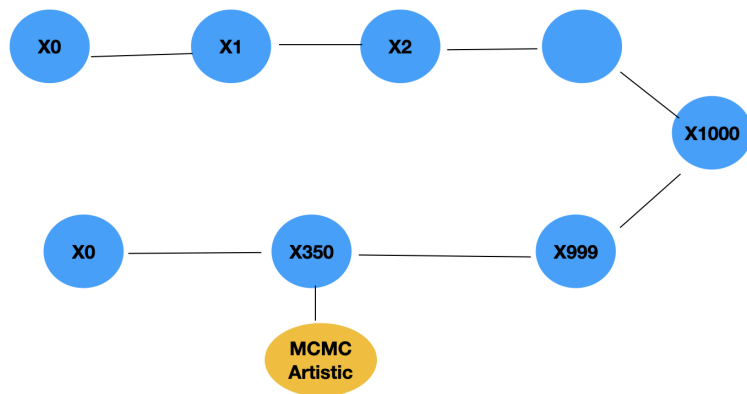
Applying sampling in the initial stages makes the noise artistic. As we move towards the 1000th step we see the samples becoming darker.

Approach 4C

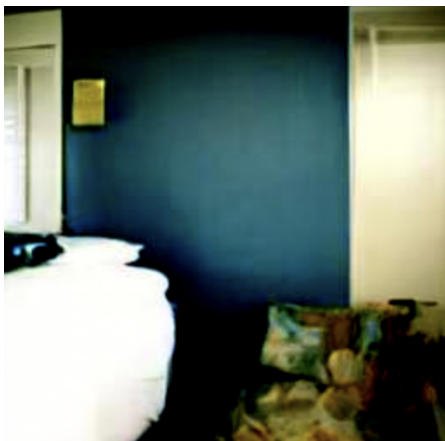
Diffusion Model + MCMC Sampling (Artistic)

Experiment Details

In this we basically think of applying MCMC only at one or a few particular Time steps and try to achieve the skew towards the desired distribution, of artistic images. This Would be possibly the most efficient approach although it demands huge hyperparameter tuning to mold out correct outputs.



Results



Original Image



Artistic Image (Band=10)

Analysis

Applying MCMC in the final steps yields a result similar to 4a)

Conclusion

Approach 1 involved fixing the function to be the one measuring the amount of redness in the image and using Langevin Metropolis Hasting Sampling at each denoising step. On some range of random seeds, the output image changes visibly, but for most of the random seeds, there is no visible change (although the score of the image increases in 3rd decimal place).

Approach 2A involved using a denoised image x_0 instead of x_t at each step for using MCMC updates. When MCMC steps were done on every denoising step, the running time was around 1 day, leading to Gaussian noise output. However, on doing MCMC steps on subsampled steps (every 100 denoising steps), the image changed to give a better score (changes in the 2nd decimal place, score maximum is 1). When the scoring function was changed to increase greenness of the image, the resulting image was the same with possibly few minor changes, suggesting that fixing the random seed fixes the output distribution such that only a few modes remain.

Approach 2B involved using a denoised image for taking the MCMC step along with Gaussian blur. This approach showed similar observations to approach 1, partly due to the fact that Gaussian blur cannot retrieve the underlying image from a very noisy image in the initial steps of the diffusion process.

Approach 3 involved using a guidance signal on the predicted mean to control the color of objects in the image. After N steps, mean and mean_diff were predicted, and the mean was adjusted using the scaling factor. In experiments where $N=500$ and scaling factor=0.001, the output adapted to the target colors, but there was a slight difference between the output and the input image.

Approach 4 involved using a diffusion model and MCMC sampling in a sequential algorithm for generating artistic images. MCMC steps were applied in three setting, one after the image was fully sampled via the reverse diffusion step. The resulting artistic image looked like an oil painting of the original image. Next, MCMC at every step, which showed poor results. Finally, MCMC before last step, which yielded result same as first setting.

References

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